

$\therefore$  The columns of  $AB$  are linearly dependent.

23) Suppose  $CA = I_n$  (the  $n \times n$  identity matrix). Show that the equation  $Ax = 0$  has only the trivial solution. Explain why  $A$  cannot have more columns than rows.

**Solution:** From the definition of the matrix multiplication method,  $A$  must have  $n$  columns. Let  $A = [a_1 \ a_2 \ \dots \ a_n]$

$$CA = [Ca_1 \ Ca_2 \ \dots \ Ca_n] \quad (\text{From the defn. of matrix multiplication})$$

**Hint:** Suppose  $x$  satisfies  $Ax = 0$ , and show that  $x$  must be  $0$ .

$$\text{Let, } Ax = 0. \quad \therefore C(Ax) = C \cdot 0 \Rightarrow (CA)x = C \cdot 0 \quad (\text{Eqn. 1})$$

$$\Rightarrow I_n x = C \cdot 0 \quad (\text{Given}) \Rightarrow I_n(x) = C \cdot (0 \cdot 0) \Rightarrow I_n x = 0 \cdot (C \cdot 0) \quad (\text{Theorem 5(b) from Chapter 1}) \Rightarrow I_n x = 0.$$

From defn. of identity matrix,  $I_n x = x. \therefore x$  must be the zero vector.

$\therefore Ax = 0$  has only the trivial solution  $\square$

From Theorem 3 of Chapter 1, the matrix equation  $Ax = 0$  has the same soln. set as the system of equations corresponding to the augmented matrix  $[A \ 0]$ . By the Row Reduction Algorithm and Theorem 1 from Chapter 1, the augmented matrix  $[A \ 0]$  can be converted to a unique rref. From Theorem 2, since the system of equations has a unique solution, there are no free variables.

By defn, each column of  $A$  must be a pivot column. A pivot position of a matrix corresponds to the leading 1 of a row in rref.  $\therefore$  The no. of rows of  $A$  must be at least as large as the no. of columns of  $A. \square$

24) Suppose  $AD = I_m$  (the  $m \times m$  identity matrix). Show that for any  $b$  in  $\mathbb{R}^m$ , the equation  $Ax = b$  has a solution. [Hint: Think about the equation  $ADb = b$ .] Explain why  $A$  cannot have more rows than columns.

**Solution:** Given,  $AD = I_m$ . Let  $b$  be an arbitrary vector in  $\mathbb{R}^m$ .  $(AD) \cdot b = I_m \cdot b \Rightarrow (AD) \cdot b = b$  (By defn. of Identity matrix)

$$\Rightarrow A(Db) = b \quad (\text{Eqn 1}) \Rightarrow Ax = b, \text{ where } x = Db$$

$\therefore x = Db$  is a solution of  $Ax = b$  for any  $b$  in  $\mathbb{R}^m$ .